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INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Department of Chemical Engineering

Mid-semester (Autumn) Examination 2016-2017

Subject: Advanced Mathematical Techniques in Chemical Engineering (CH61015)

Remarks:

1. Attempt ALL questions. All questions compulsory.
2. Time = 2 h; maximum marks = 80; total number of printed pages = 2.

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1. Determine the dimension of the equilibrium solution space of the following set of ordinary differential equations:

$$\frac{dx_1}{dt} = (cos\theta)x_1 - (sin\theta)x_2$$
$$\frac{dx_2}{dt} = (sin\theta)x_1 + (cos\theta)x_2$$
$$\frac{dx_3}{dt} = x_3$$

. . . 10 marks

2. Determine the bifurcation and sketch the phase portrait for the following family of ordinary differential equations:

$$\frac{dx}{d\theta} = asin\theta$$

. . . 10 marks

3. Sketch the straight line solutions along the eigenvectors and indicate the direction of time for the following set of ordinary differential equations:

$$\frac{dx_1}{dt} = x_1 + x_2$$
$$\frac{dx_2}{dt} = 2x_2$$
$$\frac{dx_3}{dt} = -x_2 + 4x_3$$

. . . 10 marks

4. Determine the solution vector using a suitable LU-decomposition of the following set of linear equations:

$$2x_1 - x_2 = 0$$
$$-x_1 + 2x_2 - x_3 = 0$$
$$-x_2 + 2x_3 - x_4 = 0$$
$$-x_3 + 2x_4 = 0$$

. . . 15 marks

5. A space of polynomial functions $x^n \forall n \in \mathbb{I}^+ \cup 0$ of dimension 3 needs to be expanded using a suitable orthonormal basis. If the inner product in this space is defined as $< p, q >= \int_{-1}^{1} p(x)q(x)dx$ then determine the orthonormal basis functions.

. . . 15 marks

6. Prove that the Strum-Liouville operator is a self-adjoint operator.

. . . 20 marks

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INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Department of Chemical Engineering

Mid-semester (Autumn) Examination 2022-2023

Subject: Advanced Mathematical Techniques in Chemical Engineering (CH61015)

Remarks:

1. This question paper contains two parts: **Part A** and **Part B**. Attempt both parts.
2. Write all the answers of a part together.
3. Unless otherwise stated, usual mathematical notations apply.
4. Time = 2 h; maximum marks = 60; total number of printed pages = 2.

Part A: Differential equations

1. The one dimensional transient heat transfer problem is given as:

$Bi = hL/k$

$$\rho c_p \frac{\partial T}{\partial t} = k\frac{\partial^2 T}{\partial x^2}$$

At $t = 0, T = T_0$; at $x = 0, k\frac{\partial T}{\partial x} + h(T - T_\infty) = 0$; at $x = L$, $\frac{\partial T}{\partial x} = 0$. Find the temperature profile $T(x,t)$ completely.

...10 marks

2. If $L = 5\frac{d^2}{dx^2} + x\frac{d}{dx} + 1$, at $x = 0, \frac{du}{dx} + u = 0$ and at $x = 1, \frac{du}{dx} - 3u = 0$ then find adjoint operator and boundary conditions for the adjoint problem.

...10 marks

3. Consider the ODE $y" + \lambda y = 0$ subject to the boundary conditions,
at $x = 0, \frac{dy}{dx} + \beta_1 y = 0$ and at $x = 1, \frac{dy}{dx} + \beta_2 y = 0$ where β_1, β_2 are constants. Find the eigenvalues and eigenfunctions.

...10 marks

Part B: Linear algebra

4. If V is a non-empty set such that $V = \{(x, y) : x, y \in \mathbb{R}\}$ and the following operations are defined on V:

$$\otimes : (a_1, b_1) + (a_2, b_2) = (3b_1 + 3b_2, -a_1 - a_2)$$

$$\odot : k(a_1, b_1) = (3kb_1, -ka_1)$$

$\forall (a_i, b_i) \in V, k \in \mathbb{R}$.

(a) Verify whether $\otimes$ and $\odot$ are binary operations on V.

(b) Verify whether $\otimes$ is distributive over $\odot$.

(c) Verify whether $\odot$ is distributive over $\otimes$.

. . . 4+3+3 = 10 marks

5. For the following two systems of equations in unknowns x_i's and y_i's, determine whether the corresponding range spaces are identical.

System 1

$$\begin{aligned} 2x_1 + 6x_2 + 9x_3 + 7x_4 &= p_1 \\ x_1 + 3x_2 + 3x_3 + 2x_4 &= p_2 \\ -x_1 - 3x_2 + 3x_3 + 4x_4 &= p_3 \end{aligned}$$

System 2

$$\begin{aligned} 3y_1 + y_2 + 10y_3 + 14y_4 &= q_1 \\ y_1 + 2y_3 + 3y_4 &= q_2 \\ y_2 + 4y_3 + 5y_4 &= q_3 \end{aligned}$$

. . . 10 marks

6. Determine a basis for the solution space of the following simultaneous equations:

$$\frac{dy}{dx} - 2y = 0$$

$$\frac{dy}{dx} - \frac{y}{x} = 0$$

. . . 10 marks

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Sol'n :

1. The one dimensional transient heat transfer problem is given as:

$$\rho c_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}$$

At $t = 0, T = T_0$; at $x = 0, k\frac{\partial T}{\partial x} + h(T - T_\infty) = 0$; at $x = L, \frac{\partial T}{\partial x} = 0$. Find the temperature profile $T(x,t)$ completely.

$$\rho C_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}$$

$t = 0$, $T = T_0$

$x = 0$, $k\frac{\partial T}{\partial x} + h(T - T_\infty) = 0$

$x = L$, $\frac{\partial T}{\partial x} = 0$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$\theta = \frac{T - T_\infty}{T_0 - T_\infty}$, $\alpha = \frac{k}{\rho C_p}$, $\tau = \frac{\alpha t}{L^2}$, $x^* = x/L$

$$\frac{\partial \theta}{\partial \tau} = \frac{\partial^2 \theta}{\partial x^{*2}}$$

$\tau = 0$, $\theta = \boxed{1}$

$x^* = 0$, $\frac{k(\theta(T_0 - T_\infty) + T_\infty)}{\partial L x^*} + h(\theta)(T_0 - T_\infty) = 0$

$$\frac{k}{L} \cancel{(T_0 - T_\infty)} \frac{\partial \theta}{\partial x^*} + h\theta \cancel{(T_0 - T_\infty)} = 0$$

$$\frac{\partial \theta}{\partial x^*} + \frac{hL}{k}\theta = 0$$

$$\frac{\partial \theta}{\partial x^*} + Bi\,\theta = \boxed{0}$$

$x^* = 1$, $\frac{\partial \theta}{\partial x^*} = \boxed{0}$

Now

$$\theta = T(\tau)\, X(x^*)$$

$$X \frac{\partial T}{\partial \tau} = T \frac{\partial^2 X}{\partial x^{*2}}$$

$$\frac{1}{T}\frac{\partial T}{\partial \tau} = \frac{1}{X}\frac{\partial^2 X}{\partial x^{*2}} = -\alpha^2$$

$$X = C_1 \cos(\alpha x^*) + C_2 \sin(\alpha x^*)$$

At $x^* = 1$, $\frac{\partial \theta}{\partial x^*} = 0 \Rightarrow \frac{\partial X}{\partial x^*} = 0$

$$\Rightarrow -C_1 \cancel{\alpha} \sin(\alpha) + \cancel{\alpha} C_2 \cos\alpha = 0$$

$$\Rightarrow \boxed{C_2 = C_1 \tan\alpha}$$

$$x^* = 0, \quad \frac{\partial \theta}{\partial x^*} + Bi\,\theta = 0 \Rightarrow \frac{\partial X}{\partial x^*} + Bi\,X = 0$$

$$\Rightarrow \quad -c_1 \alpha \sin(\alpha x^*) + \alpha c_2 \cos(\alpha x^*) + Bi\,c_1 \cos(\alpha x^*) + Bi\,c_2 \sin(\alpha x^*) = 0.$$

$$\Rightarrow \quad [\alpha C_2 + Bi\,C_1] \cos \alpha x^* + \cancel{[Bi\,C_2 - C_1 \alpha]} \sin \alpha x^* = 0$$

putting $x^* = 0$

$$\Rightarrow \quad [\alpha c_2 + Bi\,c_1] = 0.$$

$$\alpha c_1 \tan \alpha + Bi\,c_1 = 0$$

$$c_1 (\alpha \tan \alpha + Bi) = 0$$

Roots of $\boxed{\alpha + Bi \cot \alpha = 0}$ will be eigenvalues

$$\frac{1}{T} \frac{\partial T}{\partial \tau} = -\alpha_n^2 \Rightarrow T = c_3 \exp(-\alpha_n^2 \tau)$$

Rewriting $X \Rightarrow c_1 \cos \alpha x^* + c_2 \sin \alpha x^*$

$$\Rightarrow c_1 \cos \alpha x^* + c_1 \tan \alpha \sin \alpha x^*$$

$$\Rightarrow c_1 [\cos \alpha x^* + \tan \alpha \sin \alpha x^*]$$

$$\theta = \sum \theta_n = \sum C_n \exp(-\alpha_n^2 \tau) [\cos \alpha x^* + \tan \alpha \sin \alpha x^*]$$

$$\tau = 0 \Rightarrow \theta = 1$$

$$1 = \sum c_n [\cos \alpha x^* + \tan \alpha \sin \alpha x^*]$$

$$\int_0^1 \cos \alpha_m x^* + \tan \alpha_m \sin \alpha_m x^* \, dx^*$$

$$= c_n \int_0^1 \left[\cos^2 \alpha x^* + \tan^2 \alpha \sin^2 \alpha x^* + 2 \tan \alpha \sin \alpha x^* \cos \alpha x^*\right] dx^*$$

$$\frac{1}{\alpha_m} \left[\sin \alpha_m x^* - \tan \alpha_m \cos \alpha_m x^* \right]_0^1$$

$$= c_n \int_0^1 \left(\frac{1 + \cos 2\alpha x^*}{2} + \tan^2 \alpha \left(\frac{1 - \cos 2\alpha x^*}{2} \right) + \tan \alpha \sin 2\alpha x^* \right] dx^*$$

Solving LHS :

$$\frac{1}{\alpha_m} \left[\sin \alpha_m - \tan \alpha_m \cos \alpha_m + \tan \alpha_m \right]$$

Solving RHS :

$$c_n \left[\frac{1}{2} + \frac{1}{4\alpha} \sin 2\alpha + \tan^2 \alpha \left[\frac{1}{2} - \frac{1}{4\alpha} \sin 2\alpha \right] - \frac{\tan \alpha \left[\cos 2\alpha - 1 \right]}{2\alpha} \right]$$

Sol'n :

2. If $L = 5\frac{d^2}{dx^2} + x\frac{d}{dx} + 1$, at $x = 0$, $\frac{du}{dx} + u = 0$ and at $x = 1$, $\frac{du}{dx} - 3u = 0$ then find adjoint operator and boundary conditions for the adjoint problem.

$$L = 5 \frac{d^2}{dx^2} + x \frac{d}{dx} + 1$$

$$Lu = 5 \frac{d^2 u}{dx^2} + x \frac{du}{dx} + u$$

$$\langle v, Lu \rangle \Rightarrow \int_0^1 v \, L u \, dx$$

$$\Rightarrow \int_0^1 v \left[5u'' + x u' + u \right] dx$$

$$\Rightarrow \left[5vu'\right]_0^1 - \int_0^1 5v'u' + \int_0^1 v(xu'+u)dx$$

$$\Rightarrow \left[5vu'\right]_0^1 - \left[5v'u\right]_0^1 + \int_0^1 5v''u\,dx + \left[vux\right]_0^1 + \int_0^1 v'ux\,dx$$

$$\Rightarrow \left[5vu'\right]_0^1 - \left[5v'u\right]_0^1 + \int_0^1 \left[5v'' + v'x\right]u\,dx + \left[vux\right]_0^1$$

$$J(u,v) = \left[5vu' - 5v'u + vux\right]_0^1$$

$$\boxed{L^* = 5\frac{d^2}{dx^2} + x\frac{d}{dx}}$$

For $J(u,v) = 0$

$$\left[5v(1)u'(1) - 5v'(1)u(1) + v(1)u(1) - 5v(0)u'(0) + 5v'(0)u(0)\right]$$

boundary conditions

$$u'(0) + u(0) = 0 \Rightarrow u'(0) = -u(0)$$

$$u'(1) - 3u(1) = 0 \Rightarrow u'(1) = 3u(1)$$

$$J(u,v) = 15v(1)u(1) - 5v'(1)u(1) + v(1)u(1) + 5v(0)u(0) + 5v'(0)u(0)\Big]$$

$$\Rightarrow u_1\left[16v(1) - 5v'(1)\right] + u(0)\left[5v(0) + 5v'(0)\right] = 0$$

At $x=0$, $\underline{\underline{v + \frac{dv}{dx} = 0}}$, At $x=1$, $\underline{\underline{16v - 5\frac{dv}{dx} = 0}}$

3. Consider the ODE $y'' + \lambda y = 0$ subject to the boundary conditions, at $x = 0$, $\frac{dy}{dx} + \beta_1 y = 0$ and at $x = 1$, $\frac{dy}{dx} + \beta_2 y = 0$ where β_1, β_2 are constants. Find the eigenvalues and eigenfunctions.

Sol'n

$$\frac{d^2y}{dx^2} + \lambda y = 0 \quad , \lambda = +\alpha^2 \text{ (Non-Trivial soln)}$$

$x = 0$, $\frac{dy}{dx} + \beta_1 y = 0$

$x = 1$, $\frac{dy}{dx} + \beta_2 y = 0$

$$y = c_1 \sin \alpha x + c_2 \cos \alpha x$$

At, $x = 0$

$$c_1 \alpha \cos \alpha x - c_2 \alpha \sin \alpha x + \beta_1 c_1 \sin \alpha x + \beta_1 c_2 \cos \alpha x = 0$$

$$c_1 \alpha + \beta_1 c_2 = 0 \Rightarrow c_2 = \frac{-c_1 \alpha}{\beta_1}$$

At, $x = 1$

$$c_1 \alpha \cos \alpha x - c_2 \alpha \sin \alpha x + \beta_2 c_1 \sin \alpha x + \beta_2 c_2 \cos \alpha x = 0$$

$$(c_1 \alpha + \beta_2 c_2) \cos \alpha + (\beta_2 c_1 - c_2 \alpha) \sin \alpha = 0$$

$$\left(c_1 \alpha - \beta_2 \frac{c_1}{\beta_1} \alpha\right) \cos \alpha + \left(\beta_2 c_1 + \frac{c_1 \alpha^2}{\beta_1}\right) \sin \alpha = 0$$

$$c_1 \underbrace{\left[\alpha \cos \alpha - \frac{\beta_2}{\beta_1} \alpha \cos \alpha + \beta_2 + \frac{\alpha^2}{\beta_1} \sin \alpha \right]}_{\text{roots of this eq'n } \alpha_n \text{ are the eigenvalues}} = 0$$

$$y_n = c_1 \sin \alpha_n x - c_1 \frac{\alpha_n}{\beta_1} \cos \alpha_n x$$

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Mid-semester (Autumn) Examination 2018-2019

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Remarks:

1. This question paper contains two parts: **Part A** and **Part B**. Attempt both parts.
2. Write all the answers of a part together.
3. Unless otherwise stated, usual mathematical notations apply.
4. Time = 2 h; maximum marks = 60; total number of printed pages = 3.

Part A: Linear algebra

1. Prove that a set of vectors of the form $[x_1, x_2 \dots x_n]^T \in \mathbb{R}^n$, satisfying m equations (given below) is a subspace of $\mathbb{R}^n(\mathbb{R})$

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0$$
$$\cdots$$
$$\cdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0$$

where $a_{ij} \in \mathbb{R}\ \forall\ i, j$.

. . . 10 marks

2. For the following two systems of equations in unknowns x_i's and y_i's, determine whether the corresponding range spaces are identical.

System 1

$$x_1 + 3x_2 + 3x_3 + 2x_4 = a_1$$
$$2x_1 + 6x_2 + 9x_3 + 7x_4 = a_2$$
$$-x_1 - 3x_2 + 3x_3 + 4x_4 = a_3$$

System 2

$$y_1 + 2y_3 + 3y_4 = b_1$$
$$y_2 + 4y_3 + 5y_4 = b_2$$
$$3y_1 + y_2 + 10y_3 + 14y_4 = b_3$$

. . . 10 marks

3. Let $\underline{\underline{A}} = \begin{bmatrix} 3 & -2 \\ 4 & -5 \end{bmatrix}$ be a matrix in $\mathbb{R}^{2\times 2}$ which defines a linear transformation $t_A : \mathbb{R}^2 \to \mathbb{R}^2$ by the rule $t_A(\underline{x}) = \underline{\underline{A}}\underline{x} \ \forall \ \underline{x} \in \mathbb{R}^2$. Find the matrix of t_A relative to the basis $B = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix} \right\}$ for $\mathbb{R}^2$.

. . . 10 marks

Part B: Differential equations

4. Consider a Sturm-Liouville problem $Lu = \lambda r(x)u$ where L is the operator, λ is the eigenvalue, and $r(x)$ is the weight function. If λ_m and λ_n are distinct eigenvalues, prove that the corresponding eigenfunctions, u_m and u_n, are orthogonal functions with respect to weight function r.

. . . 10 marks

5. Solve the following problem completely including all the derivations:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

subject to boundary conditions:

at $x = 0, \dfrac{\partial u}{\partial x} = 0$

at $x = 1, u = 0$

at $y = 0, \dfrac{\partial u}{\partial y} = -q_0$

at $y = 1, \dfrac{\partial u}{\partial y} + 5u = 0$

. . . 10 marks

solve yourself

$\boxed{6.}$ Solve the following transient heat conduction problem:

$$\rho c_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}$$

At $t = 0, T = T_0(1 + x)$

At $x = 0, \dfrac{\partial T}{\partial x} = 0$

At $x = L, -k\dfrac{\partial T}{\partial x} = h(T - T_\infty)$

Make the above equation suitably non-dimensional and completely solve the problem.

...**10 marks**

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Sol'n :

5. Solve the following problem completely including all the derivations:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

subject to boundary conditions:

at $x = 0, \dfrac{\partial u}{\partial x} = 0$

at $x = 1, u = 0$

at $y = 0, \dfrac{\partial u}{\partial y} = \boxed{-q_0}$

at $y = 1, \dfrac{\partial u}{\partial y} + 5u = 0$

$u = X(x)\,Y(y)$

$Y\dfrac{d^2X}{dx^2} + X\dfrac{d^2Y}{dy^2} = 0$

$-\dfrac{1}{Y}\dfrac{d^2Y}{dy^2} = \dfrac{1}{X}\dfrac{d^2X}{dx^2} = -\alpha^2$

$X = C_n \cos \alpha_n x \quad , \alpha_n = (2n-1)\dfrac{\pi}{2}$

$Y = c_1 e^{\alpha_n y} + c_2 e^{-\alpha_n y}$

$y = 1, \quad \dfrac{\partial Y}{\partial y} + 5Y = 0$

$\Rightarrow \quad c_1 \alpha_n e^{\alpha_n} - \alpha_n c_2 e^{-\alpha_n} + 5c_1 e^{\alpha_n} + 5c_2 e^{-\alpha_n} = 0$

$c_1 e^{\alpha_n}(\alpha_n + 5) = c_2 e^{-\alpha_n}[\alpha_n - 5]$

$$\frac{c_1 e^{2\alpha_n}(\alpha_n+5)}{(\alpha_n-5)} = c_2$$

$$\Rightarrow \quad Y(y) = c_1 e^{\alpha_n y} + \Big[c_1 e^{2\alpha_n} \underbrace{\frac{(\alpha_n+5)}{(\alpha_n-5)}}_{k} \Big] e^{-\alpha_n y}$$

$$\Rightarrow \quad c_1 e^{\alpha_n y} + c_1 e^{2\alpha_n} k\, e^{-\alpha_n y}$$

$$\Rightarrow \quad c_1 e^{\alpha_n} \left[e^{-\alpha_n(1-y)} + k e^{\alpha_n(1-y)} \right]$$

$$\Rightarrow \quad u = \sum_{i=1}^{n} X(x)\, Y(y)$$

$$\Rightarrow \sum C_n e^{\alpha_n} \left[e^{-\alpha_n(1-y)} + k e^{\alpha_n(1-y)} \right] \cos \alpha_n x$$

At $y = 0$, $\dfrac{\partial u}{\partial y} = -q_0$

$$\left.\frac{\partial u}{\partial y}\right|_{y=0} = \sum C_n e^{\alpha_n} \left[\alpha_n e^{-\alpha_n} - k \alpha_n e^{\alpha_n} \right] \cos \alpha_n x = -q_0$$

$$\Rightarrow \sum C_n \alpha_n \left[1 - k e^{2\alpha_n} \right] \cos \alpha_n x = -q_0$$

$$C_n \alpha_n \left[1 - k e^{2\alpha_n}\right] \int_0^1 \cos^2 \alpha_n x \, dx = \int_0^1 -q_0 \cos \alpha_n x$$

$$C_n \alpha_n \left[1 - k e^{2\alpha_n}\right] \times \frac{1}{2} = -q_0 \frac{\sin \alpha_n}{\alpha_n}$$

$$C_n \Rightarrow \frac{-2q_0 \sin \alpha_n}{\alpha_n^2 \left[1 - k e^{2\alpha_n}\right]}$$

substitute & got it.